EMCH792 Final Project Report

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The script used 100 samples at 0.100 s and initialized the filters from the first truth state to isolate estimator behavior from startup mismatch.

The calibration variances were estimated directly from the supplied datasets. The retained values were $R_1 = 0.087247$ and $R_2 = 0.006536$. The short tuning sweep selected an initial covariance scale of 0.50 and a 99% scalar gate with threshold 6.6349.

Problem I

The first task was to estimate the two sensor variances from the calibration data and lock the deployment sample period. The calibration pass produced $R_1 = 0.087247$ for the range-like measurement and $R_2 = 0.006536$ for the cubic heading sensor. The deployment dataset remained uniformly sampled at 0.100 seconds, which allowed the rectangular covariance update required in the assignment to be implemented directly.

Problem II

The estimators used the assignment state $x = [x, y, \psi, u]^T$, steering and acceleration inputs, the fixed geometry $l_f = l_r = 1$, and the process covariance specified in the project prompt. Both filters applied the two scalar measurement updates in sequence, which made the gating logic transparent and allowed each sensor to be rejected independently. The workflow is summarized in the explanatory CeTZ diagram below.

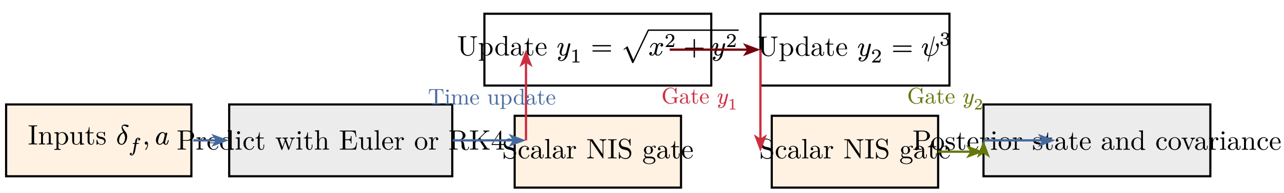

Figure 1: Sequential estimator flow used in every run. The EKF used Euler prediction only, while the UKF was evaluated with both Euler and RK4 prediction.

For the rectangular comparison without gating, the UKF was clearly more accurate than the EKF on the nonlinear trajectory. The EKF rectangular run produced a position RMSE of 3.6891 m with a heading RMSE of 17.8381 deg, while the UKF rectangular run reduced those values to 1.6879 m and 4.9532 deg. That gap is consistent with the nonlinear measurement pair, especially the ψ^3 sensor, where the sigma-point approximation handled curvature more gracefully than the first-order EKF linearization.

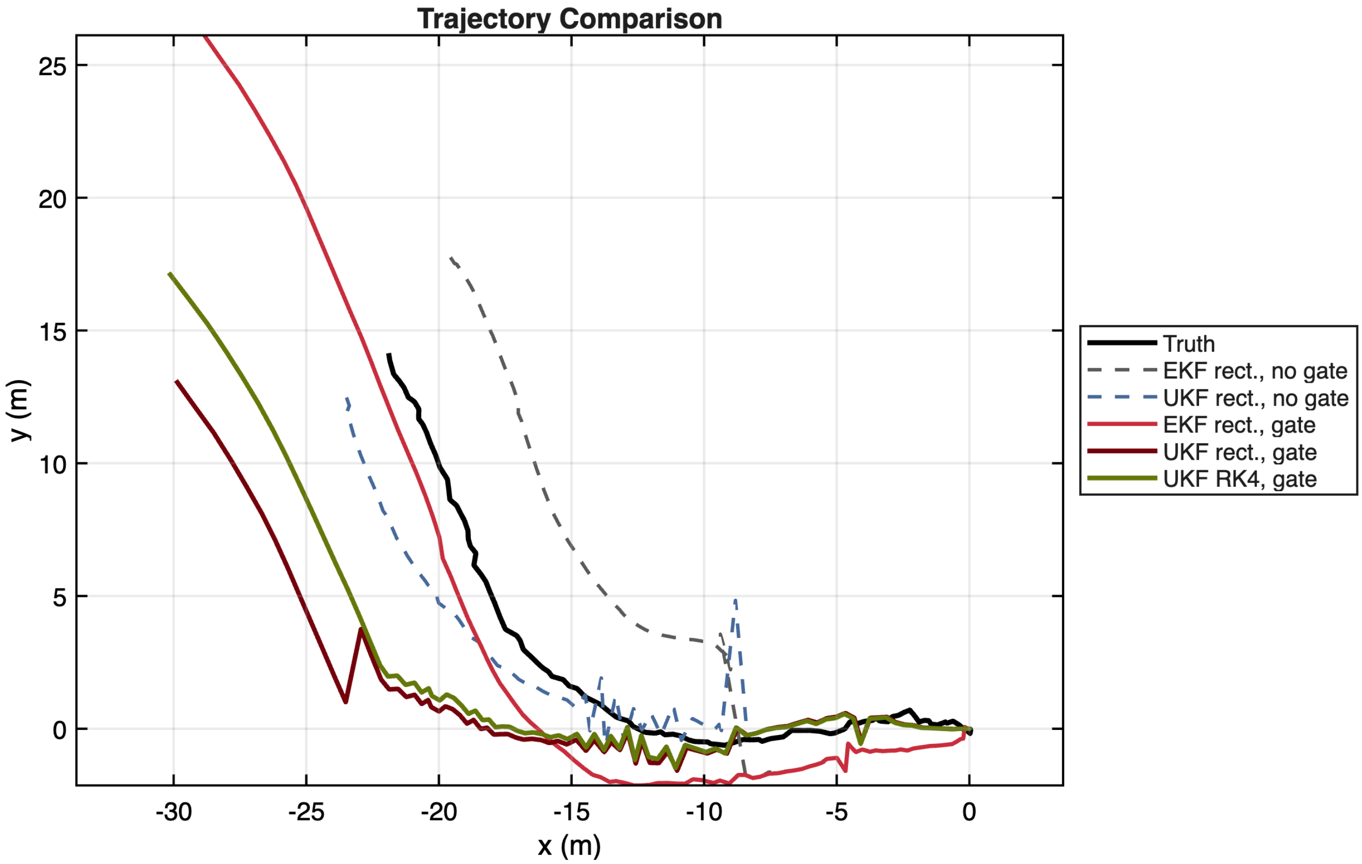

Figure 2: Truth and estimated trajectories. The no-gate UKF stays closest to the measured path, while the gated runs show the cost of relying more heavily on the imperfect process model late in the maneuver.

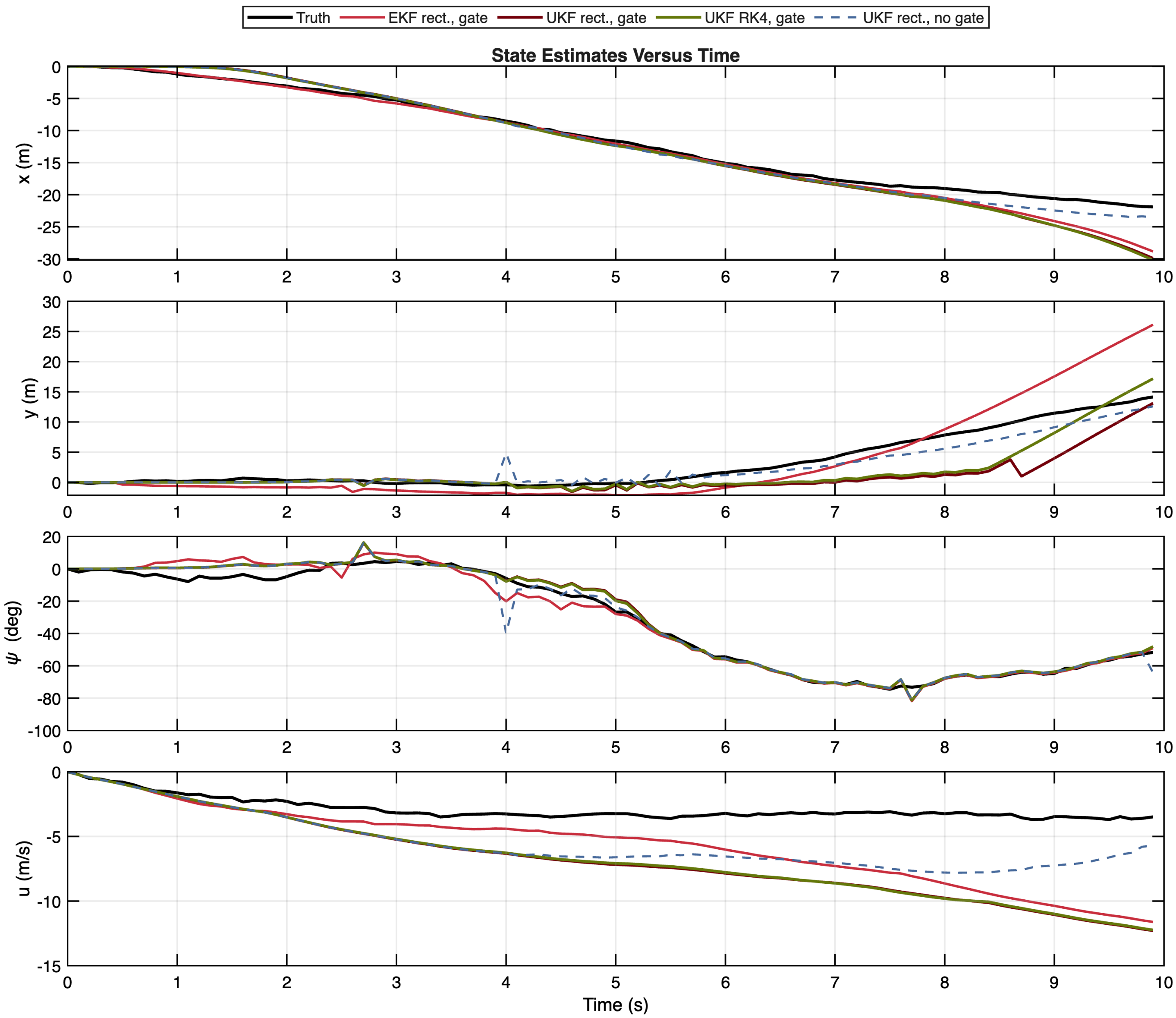

Figure 3: State histories for the principal comparison cases. The gated filters improve heading consistency but become more prediction dominated in position and speed near the end of the record.

Problem III

The tuning sweep compared two defensible scalar chi-square thresholds, namely 95% and 99%, across five initial covariance scales. The selected combination was the 99% gate with a 0.50 scaling on the base initial covariance because it minimized the composite score across the gated EKF and UKF rectangular runs.

P0 scale	Gate	Threshold	EKF pos.	UKF pos.	EKF psi deg	UKF psi deg	Score
0.50	95%	3.8415	13.6143	3.7374	8.3801	3.5998	2.4823
0.50	99%	6.6349	3.8951	4.0885	5.0388	3.8400	1.8387
1.00	95%	3.8415	16.8178	4.8416	6.1110	3.7884	2.8714
1.00	99%	6.6349	16.8178	4.5638	6.1110	3.7879	2.8548
2.00	95%	3.8415	16.6026	8.5215	4.0514	4.6785	3.0537
2.00	99%	6.6349	15.6909	6.3686	3.9223	4.7091	2.7935

P0 scale	Gate	Threshold	EKF pos.	UKF pos.	EKF psi deg	UKF psi deg	Score
4.00	95%	3.8415	16.5372	8.3965	3.4015	5.0742	3.0016
4.00	99%	6.6349	15.7881	6.5075	3.6325	5.0731	2.7646
8.00	95%	3.8415	17.3709	9.4532	4.9371	4.4273	3.0173
8.00	99%	6.6349	15.7888	5.4135	4.8713	4.4215	2.4947

The gating study showed a mixed but interpretable result. The faulty y_2 sensor was rejected three times in every final gated run, which is visible as the isolated spikes in the lower NIS plot. Those rejections reduced the heading RMSE from 17.8381 deg to 5.0388 deg for the EKF and from 4.9532 deg to 3.8400 deg for the UKF. At the same time, the rectangular gated filters also rejected a large number of y_1 measurements, namely 24 for the EKF and 14 for the UKF, so the position RMSE worsened to 3.8951 m and 4.0885 m. The practical conclusion is that the gate successfully identified the obviously bad heading-like outliers, but the process model was not strong enough to carry the full state estimate once too many range-like measurements were withheld.

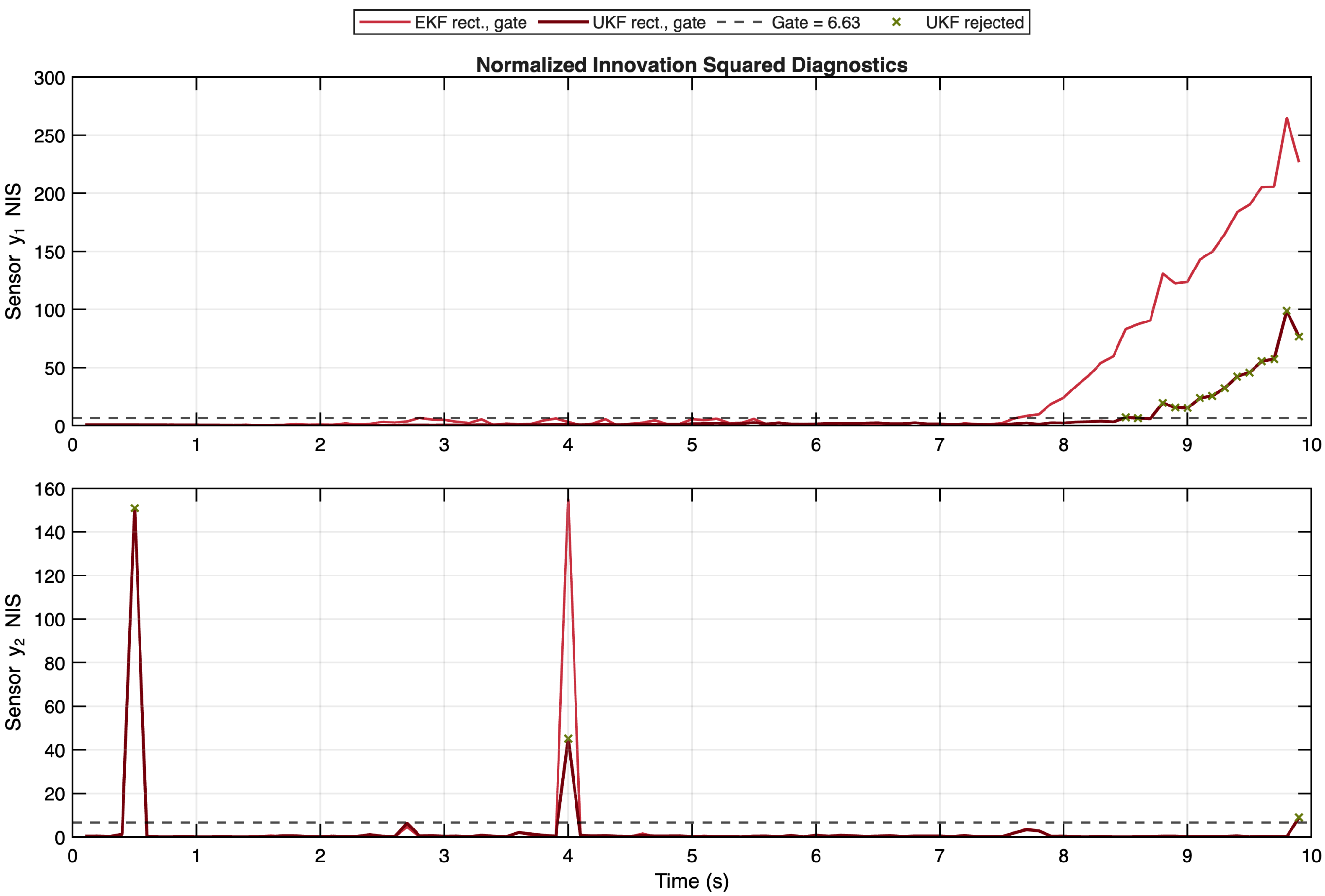

Figure 4: Normalized innovation squared histories for both sensors. The y_2 outliers are sharp and isolated, while the late y_1 residual growth explains why the gated runs become increasingly prediction dominated.

Problem IV

Replacing the UKF rectangular propagation with RK4 improved the ungated accuracy. The UKF RK4 run without gating achieved the best overall position RMSE in the study at 1.4206 m, improving on the rectangular UKF by 0.5680 m. Under the same gating logic, the RK4 variant also gave the best heading RMSE among the gated UKF runs at 3.7895 deg, although the late measurement rejections still limited the position benefit.

The covariance histories below reinforce that comparison. The UKF covariance traces are larger than the EKF trace because the sigma-point method retains more nonlinear uncertainty, and the RK4 propagation keeps that covariance evolution slightly smoother than the rectangular UKF during the final portion of the maneuver.

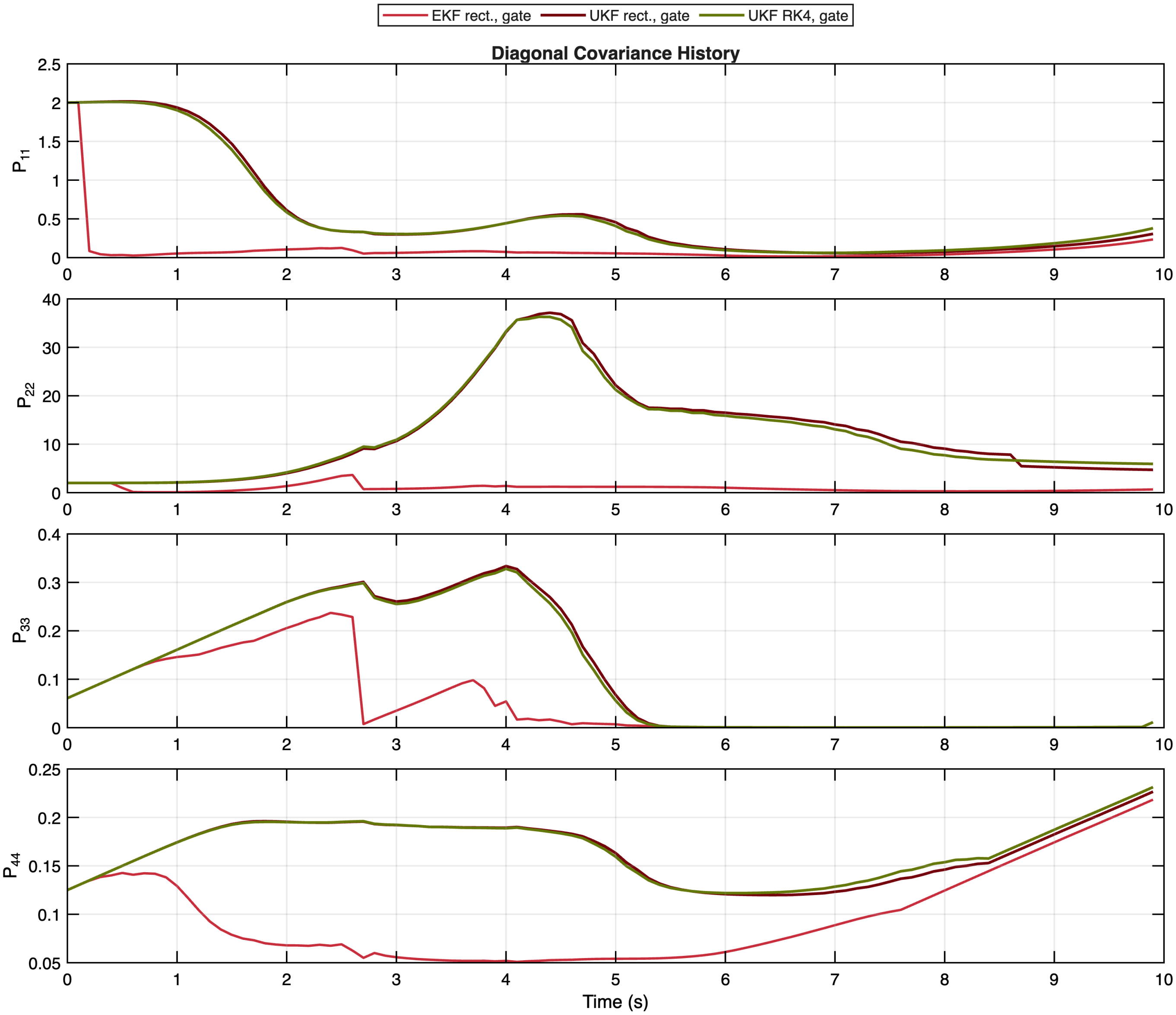

Figure 5: Diagonal covariance histories for the three gated comparison cases. The UKF variants carry more uncertainty than the EKF, which is consistent with the stronger nonlinear treatment.

Problem V

The runtime comparison used sixty repeated executions per configuration to smooth timer noise. The EKF remained the cheapest option at 2.097 ms per run on average, the rectangular UKF required 3.879 ms, and the RK4 UKF increased that cost to 4.641 ms. The RK4 propagation therefore added about 0.763 ms relative to the rectangular UKF, which is a modest penalty for a measurable gain in ungated trajectory accuracy.

Configuration	**Mean ms**	**Std ms**
EKF rect., gate	2.097	0.958
UKF rect., gate	3.879	0.157
UKF RK4, gate	4.641	0.022

Problem VI

The main conclusions are straightforward. The UKF was the better estimator for this dataset because the nonlinear dynamics and the ψ^3 measurement were strong enough for the first-order EKF approximation to lose accuracy. The covariance comparison also favored the UKF qualitatively, since the sigma-point filters carried larger and more credible uncertainty envelopes instead of the tighter EKF trace. Measurement gating helped where it needed to help, namely by removing the worst y_2 outliers and reducing heading error, but it also exposed the weakness of the process model by rejecting too many late y_1 measurements. RK4 prediction was the cleanest upgrade in the study because it produced the best overall ungated trajectory while increasing runtime only slightly.

Configuration	**x**	**y**	**psi deg**	**u**	**pos.**	**avg tr(P)**	**rej y1**	**rej y2**
EKF rect., no gate	1.1828	3.4943	17.8381	1.5688	3.6891	0.7417	0	0
UKF rect., no gate	1.0710	1.3046	4.9532	2.9613	1.6879	6.0709	0	0
EKF rect., gate	1.8305	3.4382	5.0388	3.6671	3.8951	1.1537	24	3
UKF rect., gate	2.2324	3.4252	3.8400	4.5974	4.0885	13.0001	14	3
UKF RK4, no gate	0.9634	1.0441	5.0272	2.8578	1.4206	5.9494	0	0
UKF RK4, gate	2.2670	2.6936	3.7895	4.5681	3.5206	12.7821	15	3